Beacon Network goes v2

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21st European Conference on Computational Biology

12-21 September 2022
Sitges, Barcelona

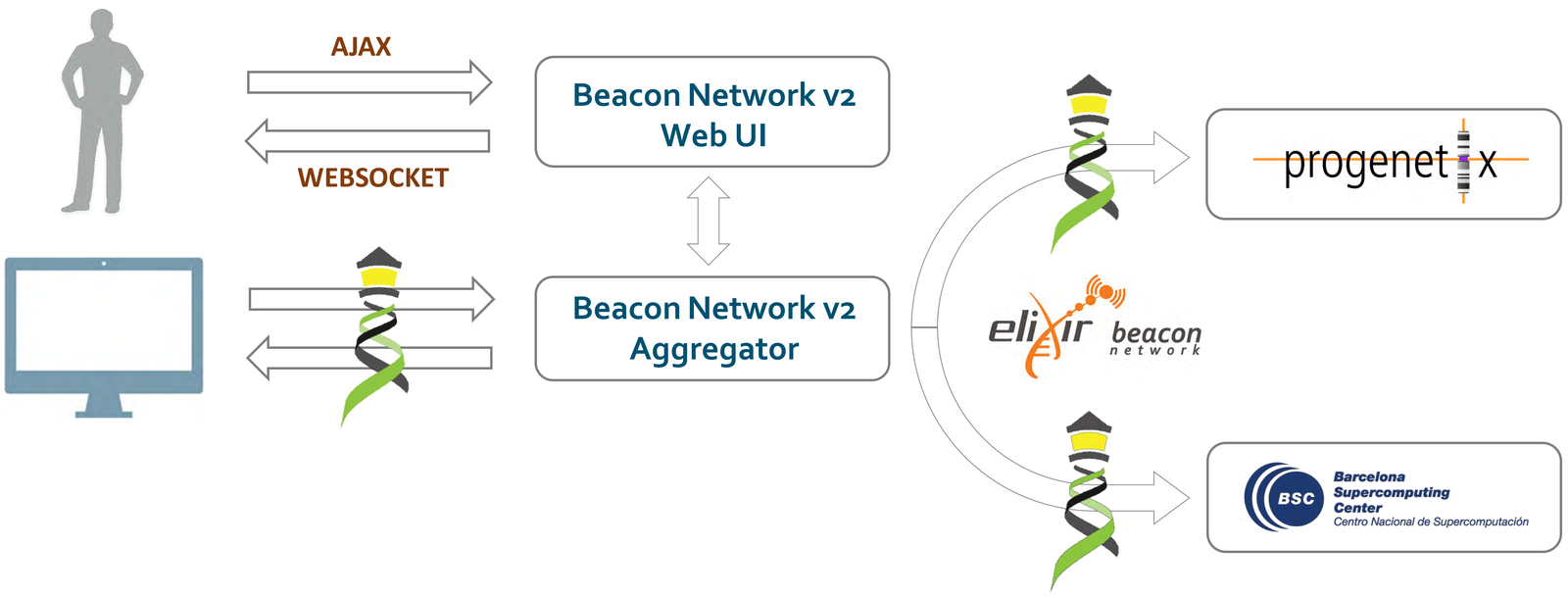
The Beacon genomic data discovery protocol has been one the most successful driver projects of the Global Alliance for Genomics and Health (GA4GH). Its original version been implemented by **hundreds of data resources**, either as standalone solutions or in networked projects with federated query concepts. Intentionally, Beacon had facilitated this through the limited scope of the protocol, allowing for an easy and - by design - privacy protecting implementation of Beacon based solutions.

After approval of the vastly extended **Beacon v2 framework** through GA4GH in 2022, current focus of the Beacon developer community is on design, testing and implementation of components and services for deployment of federated genomic services using the Beacon v2 API. In an **ELIXIR infrastructure development project** and in collaboration with international genomics and bioinformatics communities, we adhere the following approaches:

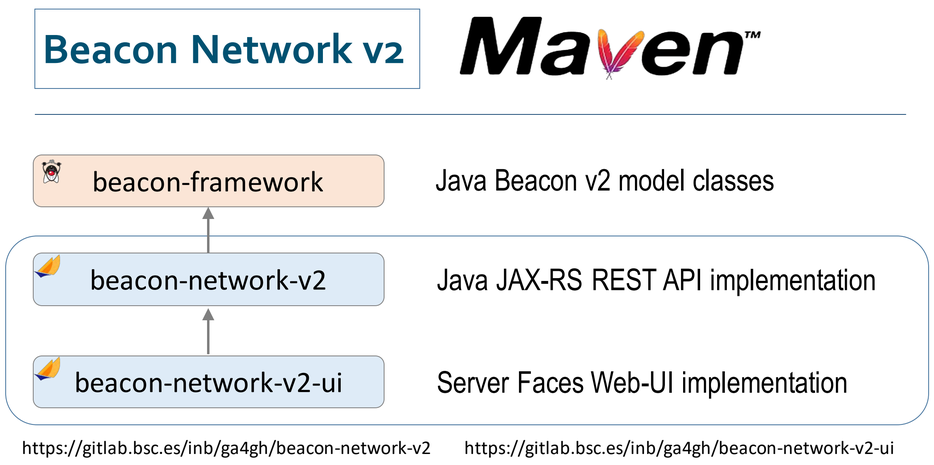
- implement networks of beacons in a controlled fashion, with close interaction between the network infrastructure developers and the participating nodes.
- develop & test services and frameworks for the automated discovery of the data types, supported query parameters and security profiles supported by individual Beacon instances, and the utilization of this information for harmonized queries and results assessment over sets of heterogeneous Beacon instances.

Design:

Beacon v2 introduced many improvements in the way its functionality is exposed to clients. These improvements greatly facilitate **Beacon Network Aggregator** which is able to introspect individual beacons in order to discover their functionality and data formats. For external clients the Aggregator looks like an ordinary Beacon providing all standard auxiliary endpoints (e.g. "info", "configuration", "map", etc.).



Implementation:



The implementation consists of two applications:

- **beacon-network-v2** is the Beacon Network v2 Aggregator that implements Beacon v2 REST API acting as a proxy distributing the queries to the network.
- **beacon-network-v2-ui** – is a user interface on top of Beacon Network APIs that add simple web interface to execute Beacon queries.
  It also includes the Aggregator so there is no need to deploy both applications.

Live Demo:

https://beacons.bsc.es/beacon-network

There is a Docker Compose demo that deploys beacon-network-v2-ui application.

The application is downloaded from the BSC GitLab maven repository and automatically updated as part of Continuous Integration (CI) process.

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